Artificial Intelligence

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Background

- Kaiqi Zhao
  - Full Professor, Harbin Institute of Technology, Shenzhen
  - Home page: https://kaiqizhao.github.io/
  - DBLP: https://dblp.org/pid/127/6172-1.html
  - Email: zhaokaiqi@hit.edu.cn

- Prior to HIT-SZ
  - Senior Lecturer, University of Auckland (QS 65)
- Education:
  - PhD - Nanyang Technological University
  - MSc - Shanghai Jiao Tong University
- Current research interests:
  - Efficient and effective methods for mining geo-spatial data
  - Probabilistic modelling, generative models
  - Graph representation learning



Purpose of this course

The goal of this course is to present cornerstone in AI, especially in the utilisation of knowledge and reasoning:

- How to solve problems via searching?
- How knowledge can be represented?
- How knowledge can be reasoned?
- How to handle uncertainty?
- How to learn knowledge from observations?

Course Contents

Themes:

- Problem solving: search, heuristics, reasoning
- Representation: search space, knowledge
- Uncertainty: randomness, probability

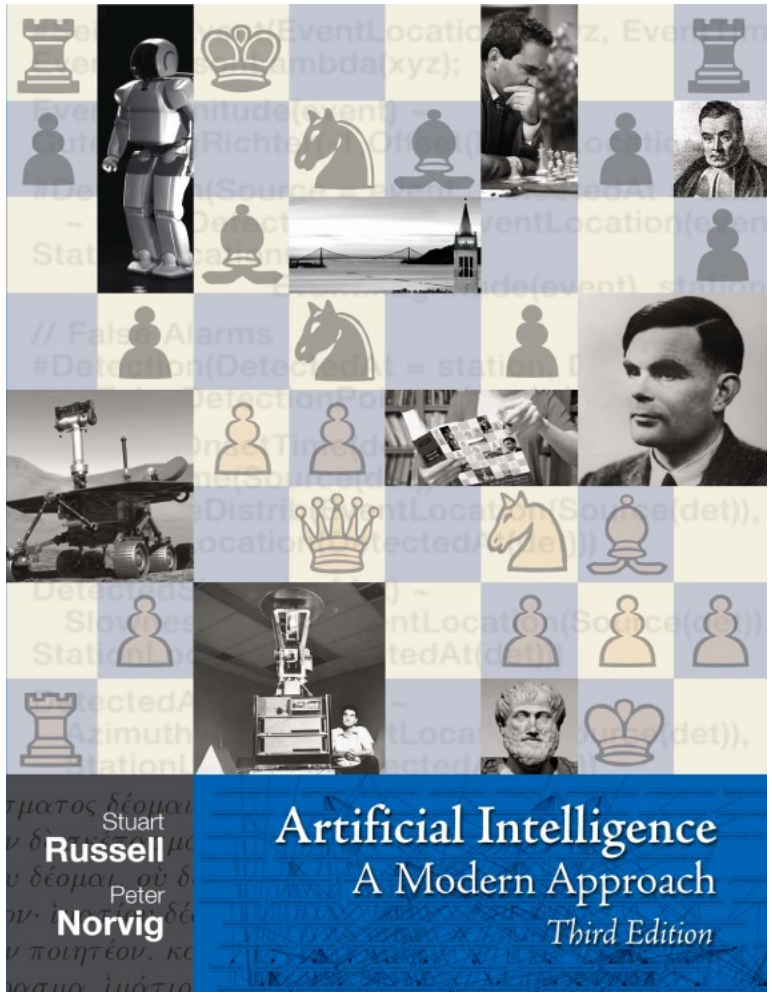
Contents:

- Search (systematical and local search, adversarial search)
- Knowledge representation
- Deterministic reasoning (logical inference)
- Reasoning with uncertainty
- Artificial neural networks

Course Information

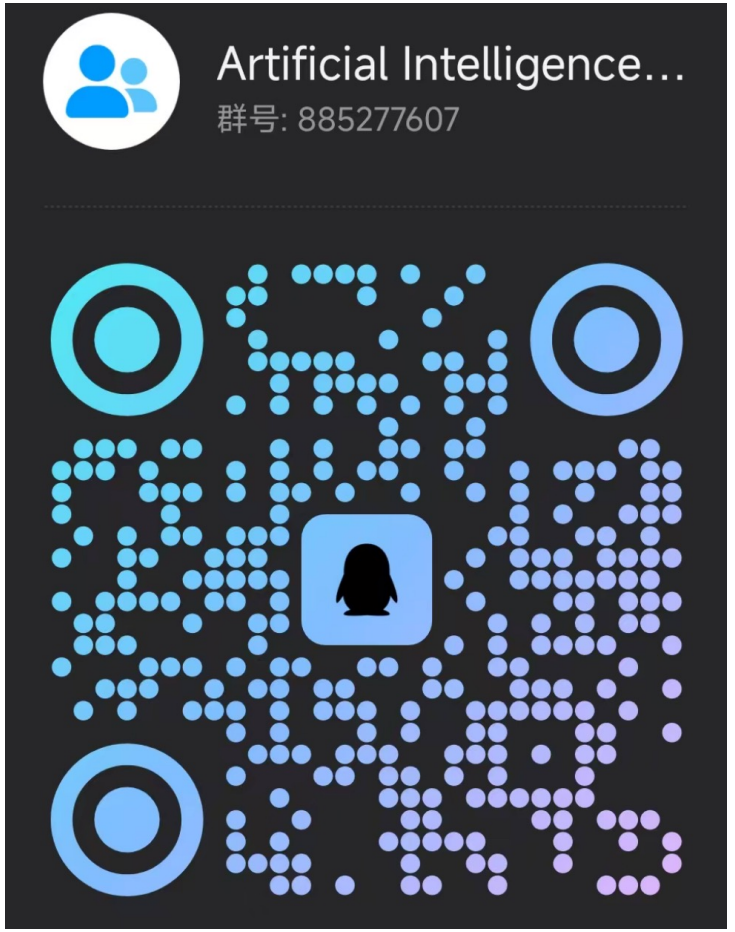
- Textbook:
  - S. Russell and P. Norvig. *Artificial Intelligence:A Modern Approach.* Third Edition or Fourth Edition

- Resources:
  - AIMA http://aima.cs.berkeley.edu/instructors.html



Assessment

- Written Exam (50%)
  - Theoretical questions
    - Understanding and insight
  - Small practical tasks
    - E.g. logic inference
- Quiz + Assignments (30%)
  - In-lecture quiz: Multiple choice
  - Assignments: Short answer
- Course Project (20%)
  - Project report



QQ Group for the course

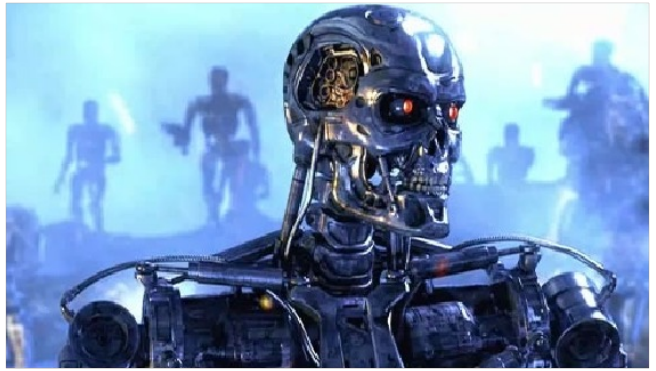
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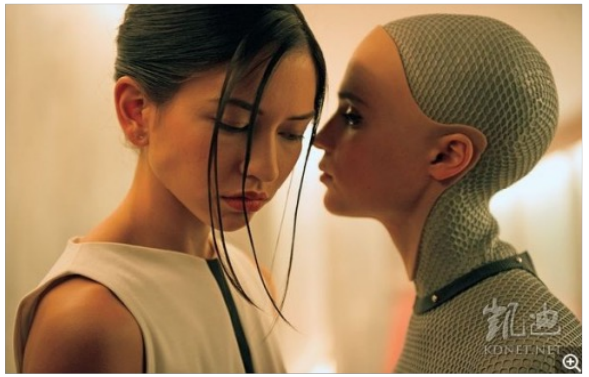
Introduction to Artificial Intelligence

What is AI?

AI in Sci-Fi









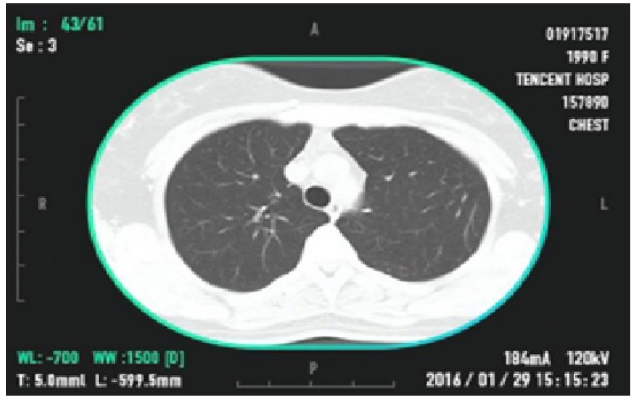


What is AI?

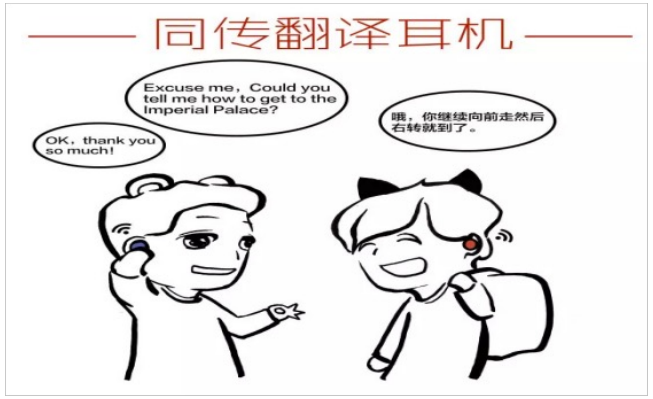
AI in real-life











What is AI?

Different definitions due to different criteria. Two dimensions:

- Thought processes/reasoning vs. behavior/action
- Success according to human standards vs. success according to an ideal concept of intelligence: rationality.

Systems that think like humans	Systems that think rationally
Systems that act like humans	Systems that act rationally

Thinking humanly

The Cognitive Science Approach

- **Goal:** To build systems that mimic the actual thought processes of the human mind.
- **Key Idea:** If we understand how a human arrives at a conclusion (the mental steps), we can program a computer to follow the same steps.

- **Key Example: Cognitive Modeling.**
  - Using insights from psychology and neuroscience to simulate human problem-solving.
  - Example: Creating a program that learns a new language by replicating the way a child's brain learns, including making the same types of mistakes.
- **Challenge**: How do we know what really goes on inside a human mind?

Acting humanly

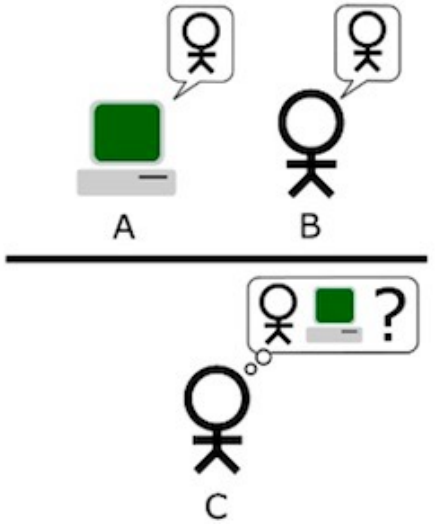
The Turing Test Approach

- **Goal:** To build systems whose behavior is indistinguishable from that of a human.
- **Key Idea:** If an AI can converse with a human and the human cannot tell they are talking to a machine, the AI has achieved human-level performance.

- **Key Example: The Turing Test & Chatbots.**
  - Q: Please write me a poet on the subject of the Forth Bridge.
  - A: Count me out on this one. I never could write poetry.
  - Q: Add 34957 to 70764.
  - A: (Pause about 30 seconds and then give as answer) 105621.



"The imitation Game (2015)"



Turing Test

Thinking rationally

The “Laws of Thought” Approach

- **Goal:** To build systems that solve problems using logical deduction and irrefutable rules.
- **Key Idea:** The world can be represented by formal symbols and logical statements. Thinking "correctly" means applying the rules of logic to these statements to reach a valid conclusion.

- **Key Example: Expert Systems & Theorem Provers.**
  - "Socrates is a man. All men are mortal. Therefore, Socrates is mortal."
  - An AI follows this formal logic to guarantee a correct conclusion.
- **Challenge:** Translating unstructured knowledge into precise logical notation is difficult.

Acting rationally

The Rational Agent Approach

- **Goal:** To build systems that act to achieve the best possible outcome (or the best expected outcome when uncertain).
- **Key Idea:** An AI is an "agent" that perceives its environment and takes actions to succeed at a given goal. It doesn't need to mimic humans or following logic—it's about doing the right thing.

- **Key Example: Modern AI Systems.**
  - **Self-driving cars:** The goal is to reach the destination safely and quickly. The car acts rationally by braking for pedestrians (even if a human might not always do so).
  - **Recommendation algorithms:** Their goal is to maximize your engagement by suggesting videos you're most likely to watch.

What is AI? – A Brief Summary

- **From a Capability Perspective:** AI refers to intelligence realized in machines (computers). It encompasses the functions typically associated with human intelligence that are performed by intelligent machines, such as judgment, reasoning, proof, recognition, perception, comprehension, design, thinking, planning, learning, and problem-solving activities.

- **From a Disciplinary Perspective:** Artificial intelligence is a discipline that studies how to construct intelligent machines or intelligent systems to simulate, extend, and expand human intelligence. It is a branch of computer science involved in the research, design, and application of intelligent machines.

Notable AI moments (2000-2024)

2003	Very large datasets: genomic sequences
2007	Very large datasets: WAC (web as corpus)
2009	Very large datasets: ImageNet
2011	IBM Watson wins Jeopardy
2012	US state of Nevada permits driverless cars
2017	Google AlphaGo beat the world's best Go player, Ke Jie AlphaZero learns boardgames by itself and beats the best programs
2020	GPT-3 performs a variety of NLP tasks without retraining
2022	ChatGPT launched by OpenAI
2024	GPT-4o handles multi-modal data OpenAI o1 launched, which achieved an 83% success rate on the American Invitational Mathematics Examination DeepSeek launched

AI research methods

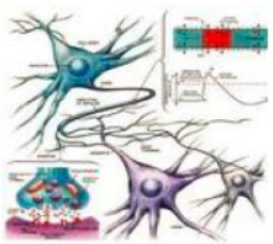
Symbolism

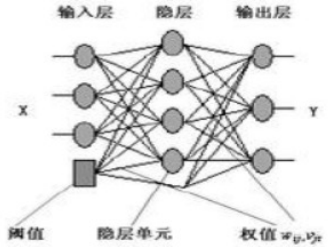
The foundation of intelligence is knowledge, and its core lies in knowledge representation and knowledge reasoning. Knowledge can be represented by symbols and can also be reasoned with using symbols, thereby enabling the establishment of a unified theoretical system for human and machine intelligence based on knowledge.



Connectionism

It proposes that the fundamental units of intelligent behavior are neurons, not symbols; the process of thinking is a process of neural connections and activities, rather than a process of symbolic computation. It opposes the Symbolist hypothesis regarding physical symbol systems.



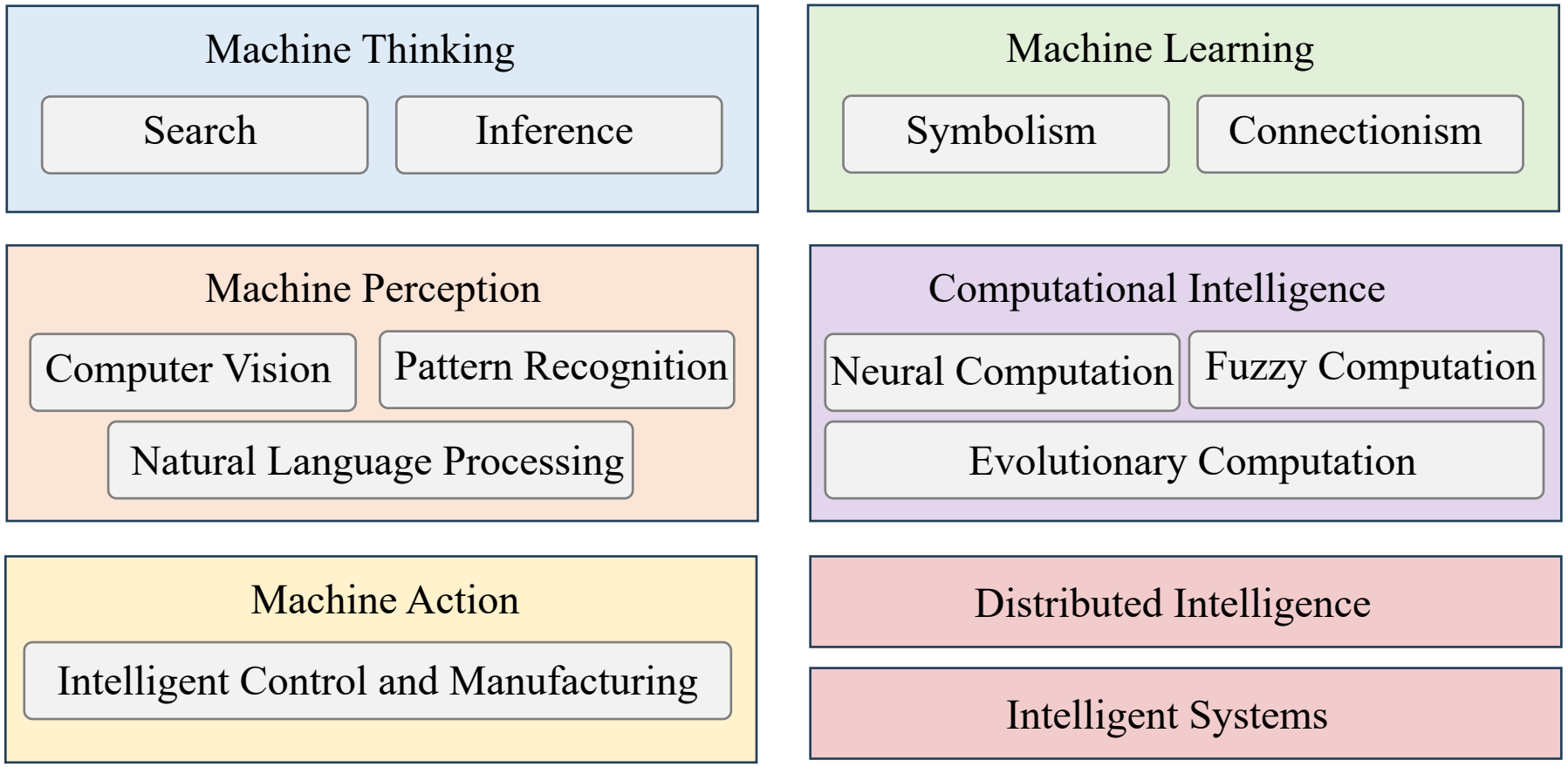


Actionism

Intelligence depends on perception and action, following a "perception-action" model. Intelligence requires no knowledge, no representation, and no reasoning. Artificial intelligence can evolve progressively, much like human intelligence.



AI research areas



Machine Thinking

Inference

- The process of starting from known facts, applying knowledge based on a certain strategy, and deriving the required conclusions.

Types of Inference: Based on the certainty of the knowledge used, it can be divided into Deterministic Reasoning and Uncertain Reasoning.

Deterministic Reasoning

- The knowledge used in the reasoning process and the conclusions drawn can be precisely represented. Their truth value is either true or false.

Uncertain Reasoning

- The knowledge used in the reasoning process and the conclusions drawn can be uncertain. The term “uncertainty” is a general designation for imprecision, fuzziness, and incompleteness.

Machine Thinking

Inference

Theoretical Foundation of Inference: Logic is a discipline that studies the laws of human thought. Mathematical logic uses mathematical methods to study logical problems.

- Deterministic reasoning is mainly based on first-order logic. It can only solve very limited problems.
- Uncertain reasoning is primarily based on non-classical logic, probability theory, and other methods.

Machine Thinking

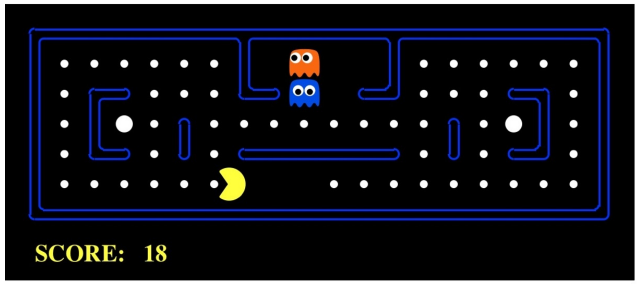
Search

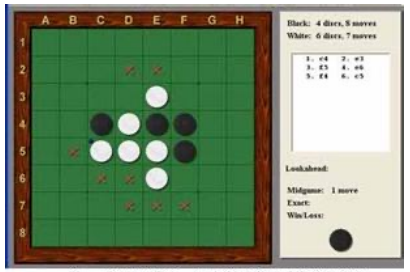
- Search refers to the process of continuously seeking reasoning paths in order to guide and control reasoning, aiming to achieve a specific goal and ultimately solve a problem.

Types of Search (Based on the representation of the problem, it can be divided into two main types):

- State-Space Search: A search method used when solving problems using the state-space representation.
- AND/OR Tree Search: A search method used when solving problems using problem reduction.

Game Tree Search: Game playing is a classic search problem. To date, research on games has primarily focused on board games like chess as the main subject.





图一 黑白棋的某一状态图（现在由黑棋下子）

Machine Perception

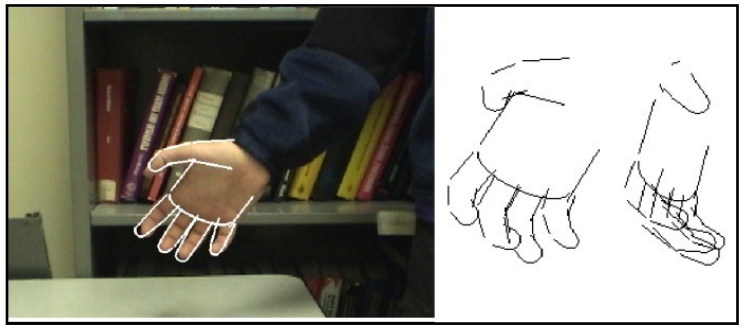
Computer Vision

- The use of computers to realize or simulate the functions of human vision. Its main research goal is to enable computers to have the ability to recognize three-dimensional environmental information through two-dimensional images.

Importance: More than 80% of the external information perceived by humans is obtained through vision.

Computer Vision refers not only to the reception of light signals but encompasses the entire process of visual information, including its acquisition, transmission, processing, storage, and understanding.



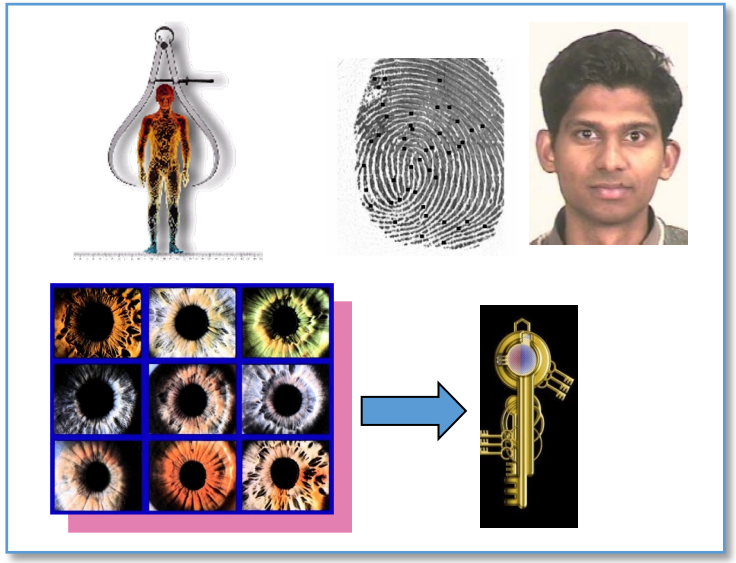


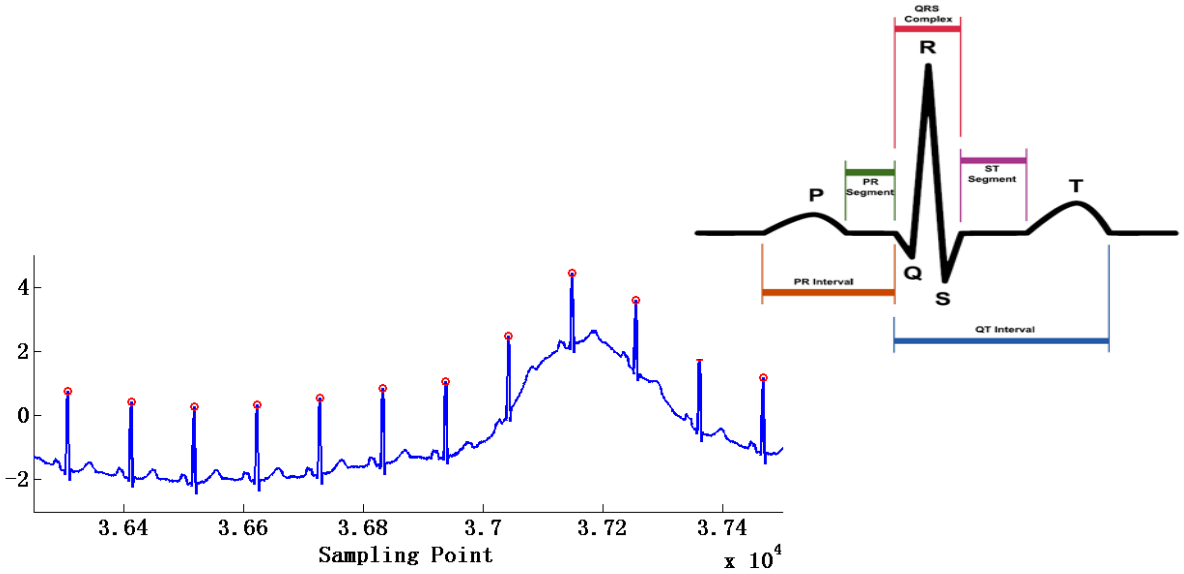


Machine Perception

Pattern Recognition

- It refers to enabling a computer to distinguish a given object and classify it into a pattern that is identical or similar to it.
- The objects to be distinguished can be physical, chemical, or physiological entities, as well as text, images, sounds, and so on.





Machine Perception

Natural Language Processing

Natural Language Processing (NLP) is the study of various theories and methods to facilitate effective communication between humans and computers.

Speech Processing



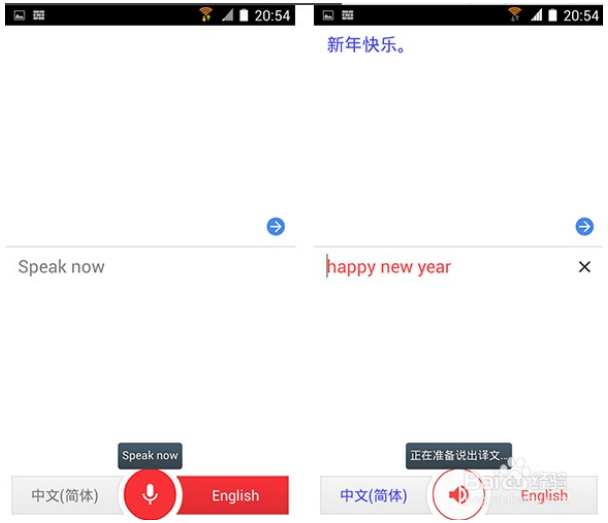


Natural Language Understanding



"Within five years, search engine will be based on an understanding of natural language."

Machine Translation

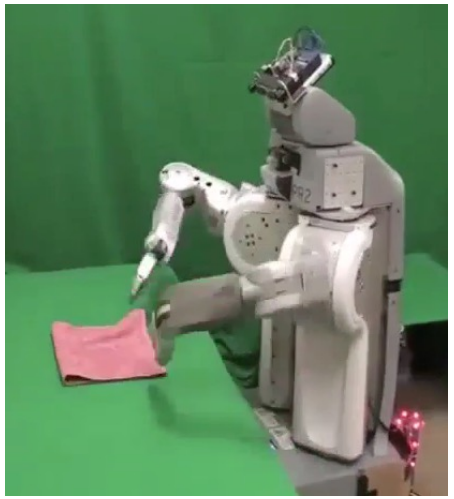


Machine Behavior

Intelligent Control/Manufacturing

- It refers to a control process that enables intelligent machines to operate independently and achieve their objectives with minimal or no human intervention. It is the product of combining artificial intelligence technology with traditional automatic control techniques.

Main Application Areas:









Machine Learning

- Machine learning enables computers to automatically acquire new knowledge like humans do, and continuously improve themselves and enhance their capabilities through practice.

- Machine learning is the fundamental approach for machines to acquire knowledge, and it is also an important indicator of machine intelligence.

- There are various methods for classifying machine learning. According to the way it simulates human learning, machine learning can be categorized into symbolic learning, neural learning, etc.

Machine Learning

Symbolic Learning

- Symbolic learning refers to machine learning methods that functionally simulate human learning abilities.

Types

- Inductive learning, deductive learning, and others.

Symbolism

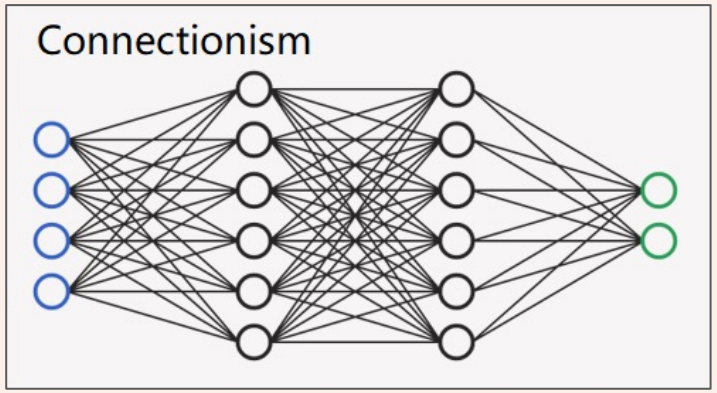
$$(p \rightarrow q) \Leftrightarrow (\neg p \vee q)$$
$$\Leftrightarrow (q \vee \neg p)$$
$$\Leftrightarrow (\neg\neg q \vee \neg p)$$
$$\Leftrightarrow (\neg q \rightarrow \neg p)$$

Neural Learning

- Neural learning, also known as connectionist learning, is a learning method based on artificial neural networks.

Types

- Perceptron learning, backpropagation (BP) network learning, Hopfield networks



Computational Intelligence

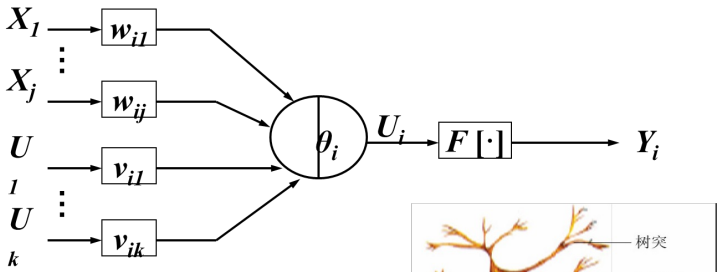
- **Computational Intelligence (CI)** draws inspiration from bionics and is based on the understanding of biological intelligence mechanisms. It employs numerical computation methods to simulate and realize human intelligence.

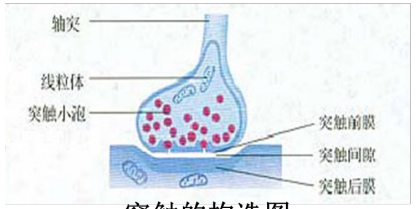
- The three fundamental areas of computational intelligence include neural computation, evolutionary computation, and fuzzy computation.

Computational Intelligence

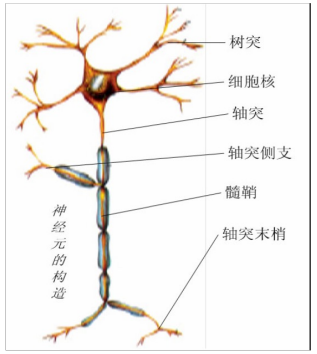
Neural Computation

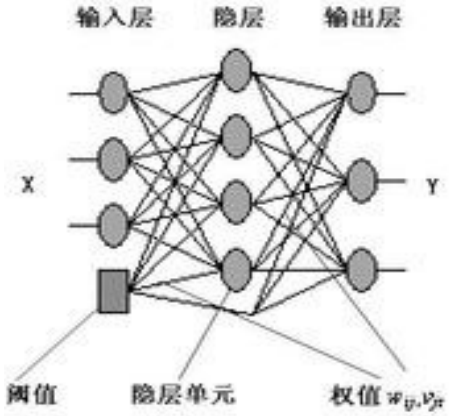
- Also known as a Neural Network (NN), it is an artificial network system formed by extensively interconnecting a large number of artificial neurons in parallel, used to simulate the structure and function of biological nervous systems.
- Neural networks possess capabilities such as self-learning, self-organization, self-adaptation, association, and fuzzy reasoning.





突触的构造图





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Computational Intelligence

Evolutionary Computation

- Evolutionary computation is inspired by Mendel's theory of genetic variation, and introduces the processes of reproduction, mutation, competition, and selection from biological evolution into algorithms. It serves as a method for simulating the evolution of human intelligence.

- **Four Main Branches of Evolutionary Computation**: Genetic algorithms, evolutionary strategies, evolutionary programming, and genetic programming.

- Genetic algorithm is the initially formed and widely influential simulation optimization algorithm within evolutionary computation.

Computational Intelligence

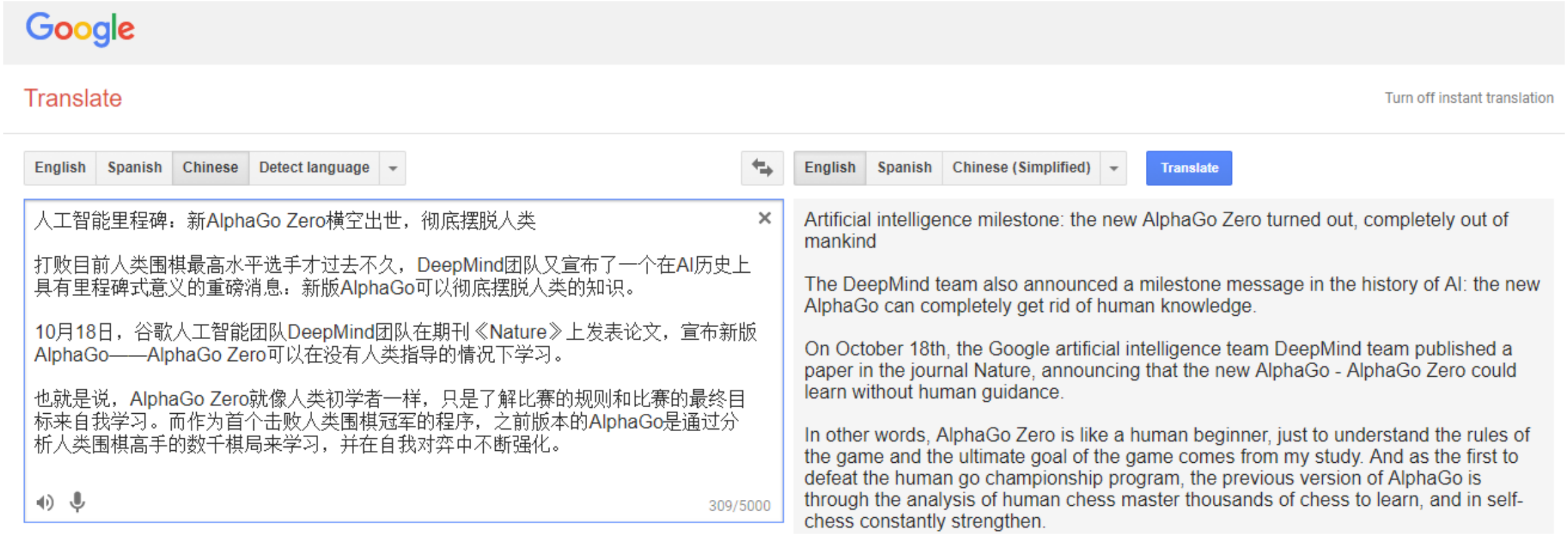
Fuzzy Computation

- Also known as fuzzy systems, it simulates human behavior using fuzzy sets and fuzzy logic, based on an understanding of the human cognitive ability to handle fuzzy phenomena.

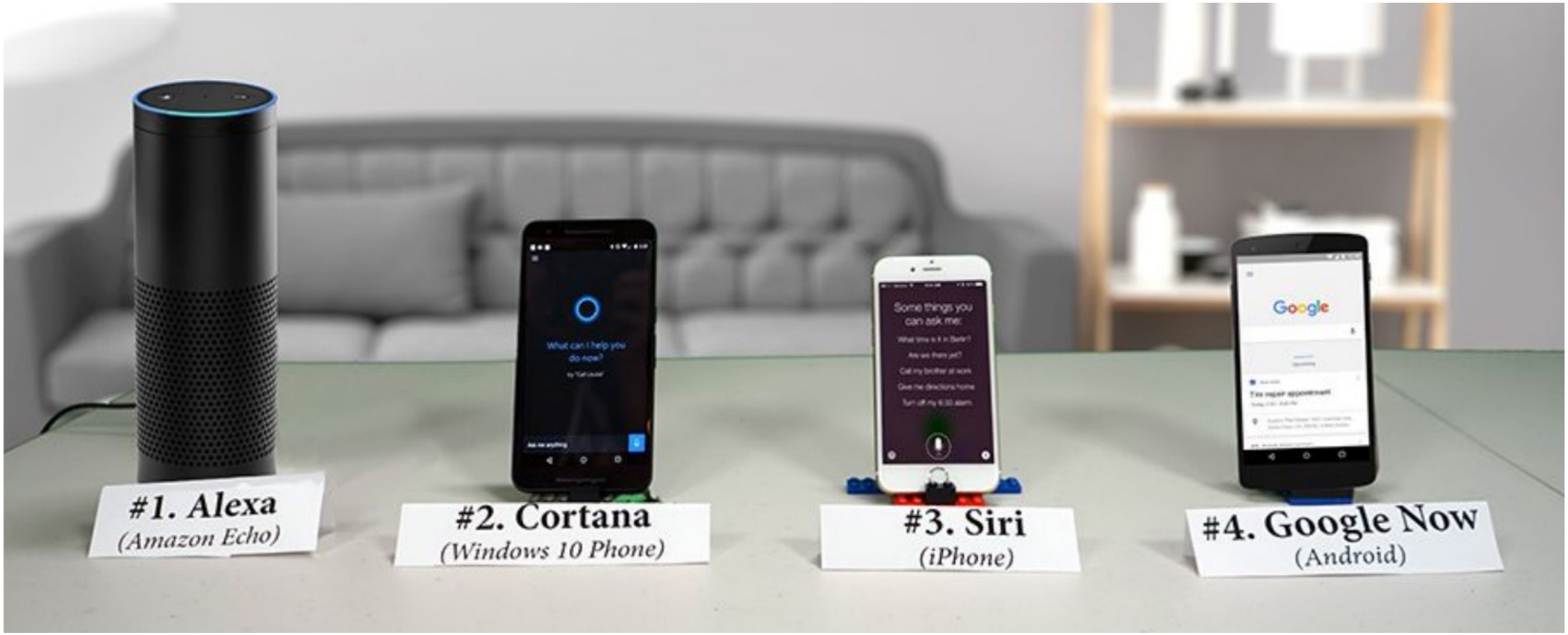
Fuzzy Concept

- It describes phenomena that cannot be precisely described due to the lack of strict boundaries. E.g. “tall”, “big”, “small” and “many”.
- Fuzzy concepts are typically represented using fuzzy sets, which allows elements to have degrees of membership, ranging between 0 and 1. E.g., Tall: {170cm → 0.7, 185cm → 0.95}

AI Applications - Machine translation



AI Applications - Speech recognition

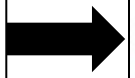


[Examples from Stanford CS221]

AI Applications - Reading comprehension

SQuAD - Oneof the most popular dataset for reading comprehension, has 100K questions taken from Wikipedia.

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".



What causes precipitation to fall?
gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?
graupel

Where do water droplets collide with ice crystals to form precipitation?
within a cloud

[Examples from Stanford CS221]

AI Applications - Autonomous driving

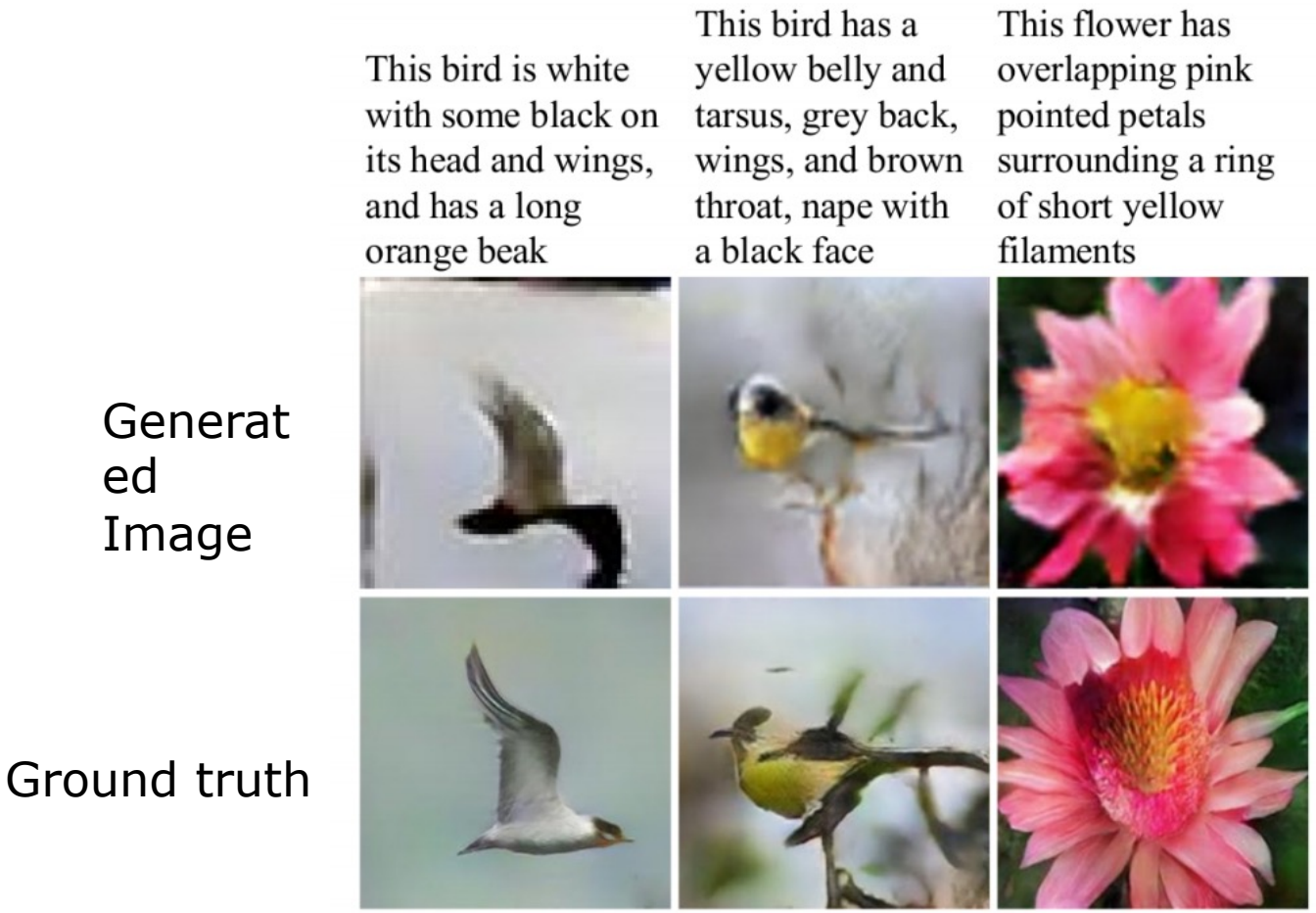
- Research in autonomous cars started in the 1980s
- The first significant event was the 2005 DARPA Grand Challenge
- 2007, DARPA created an even harder Urban Challenge, which was won by CMU
- 2009, Google started a self-driving car program
- 2015, Uber hired about 50 people from CMU's robotics department to build self-driving cars.



[Examples from Stanford CS221]

AI Applications - Image Generation

- Text → Image



[Examples from Stanford CS221]

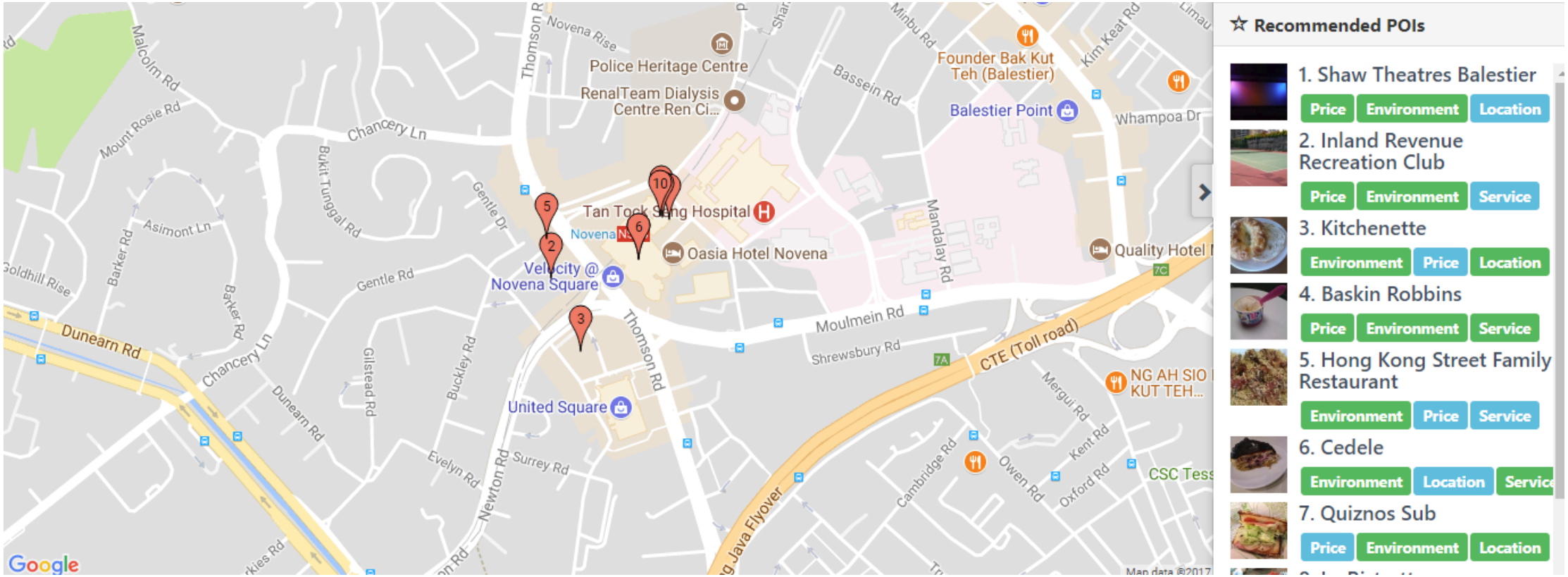
AI Applications - Artistic style transfer





AI Applications - Recommendation

- Recommend venues based on user's interests



Summary

- What is AI?
  - Two dimensions: (1) human-like v.s. rational; (2) thinking v.s. acting
- AI moments
- AI research methods
  - Symbolism, Connectionism, Actionism
- AI research areas
  - Machine thinking, machine perception, machine learning, etc.
- AI applications